

# Jesus Faustino Hernandez Conde

Physicist | Software Developer | AI & QA Specialist  
Nuevo Leon, Mexico | Bilingual: English - Spanish

## Professional Profile

Interdisciplinary professional with a background in Theoretical Physics and a double technical degree in Informatics and Computer Science. Specialized in bridging rigorous mathematical analysis with technological development. Extensive experience in Artificial Intelligence model evaluation, process automation, robotics, and software development. Capable of leading complex projects requiring a deep understanding of neural networks, application deployment, and hardware optimization.

## Professional Experience

- Artificial Intelligence Analyst (QA & Evaluation)***Data Annotation, Outlier, TRS, Alignerr*
- Comprehensive technical evaluation of responses generated by AI models (LLMs) to ensure logical, mathematical, and programming accuracy.
  - Complex dataset annotation and code development to optimize the performance and security of machine learning models in bilingual environments.
- QA Specialist for Process Automation***ByteDance*
- Execution of quality assurance (QA) tests on automated workflows focused on economic processes.
  - Systems auditing to guarantee integrity, efficiency, and error reduction in financial transactions and metrics.
- Artificial Intelligence Analyst***Google / Truelancer*
- Analysis and evaluation of AI models oriented towards email management, classification, and processing, enhancing the contextual understanding of the algorithm.
- Laboratory Administration and Technical Support***FCFM UANL*
- Diagnosis, maintenance, and component-level repair of precision laboratory equipment (GW Instek analog oscilloscopes, multimeters, power supplies).
  - Inventory management, equipment loan administration, and technical/logistical support for the Physics Olympiad.

## Notable Projects

- **Wireless Robotics with AI:** Design and programming of a wireless robotic automation system controlled via an ESP32 board. Trained a machine learning model from scratch for sign language detection and facial recognition through video transmission.
- **Bio-image Analysis and Segmentation:** Development of Python scripts (Cellpose, OpenCV, LOCO-EFA) for automatic cellular mask extraction. Applied theoretical foundations of CNN architectures and Fourier analysis.
- **Automated Laboratory Inventory System:** Development of a mobile application for university equipment management. Frontend built with React Native and Expo, backend supported by a relational database in AWS RDS, featuring barcode scanning.
- **Quantum Field Theory (QFT) Research:** Thesis focused on 1-loop ultraviolet divergence renormalization in quantum electrodynamics using FeynCalc and LaTeX.
- **Game Balancing Engine:** Creation of a simulation tool in Python to audit statistical metrics and strategic balances in Clash Royale, integrating version control with Git.

## Technical Skills

- **Languages:** C, C++, Python, R, Fortran.
- **AI & ML:** Neural Network Theory (CNNs), LLM evaluation, Cellpose, OpenCV.
- **Development & Web:** React Native, Expo, application deployment, proportional cybersecurity.
- **Data Science:** NumPy, Pandas, SciPy, Matplotlib, SymPy, FeynCalc.
- **Infrastructure & OS:** Linux, Oracle Cloud (OCI), AWS RDS, Google Colab.
- **Hardware:** Microcontrollers (ESP32), circuit and laboratory equipment repair.
- **Tools:** Git, GitHub, LaTeX.

## Education

- Bachelor of Science in Physics***Universidad Autónoma de Nuevo León (UANL) - FCFM*
- Double Technical Degree:** Informatics and Computer Science